

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: QUANTUM, ATOMIC AND MOLECULAR PHYSICS

TRINITY TERM 2015

Thursday, 18 June, 2.30 pm – 4.30 pm

10 minutes reading time

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. Describe what is meant by the *central field approximation* in multi-electron atoms, and demonstrate how it leads to the concept of electron configurations. Explain what is meant by a *term* and a *level*, and describe the interactions which lead to them. [8]

Explain what is meant by *LS coupling* and derive the Landé interval rule. [6]

One of the excited configurations of Sr is $5s5l$. This configuration gives rise to four levels, three of which have energies of 35 007, 35 022 and 35 045 cm^{-1} above the $5s^2$ ground state. Determine l , and assign L , S , and J quantum numbers to each of the four levels. [5]

Optical transitions are observed between the three levels for which energies are given in the above configuration and some of the levels in another configuration, $5s5l'$. The energies of the transitions are 20 108, 20 123, 20 146, 20 503, 20 518, and 20 689 cm^{-1} . Determine l' , and the values of L , S , and J in the upper and lower levels of each of the six transitions. [6]

2. When considering the effect of an externally applied magnetic field $\mathbf{B}_{\text{ext}}$ on the hyperfine structure of an atom with total orbital angular momentum $\mathbf{J}$ and nuclear spin $\mathbf{I}$, we employ a term in the Hamiltonian of the form

$$A_J \mathbf{I} \cdot \mathbf{J} + g_J \mu_B \mathbf{J} \cdot \mathbf{B}_{\text{ext}} - g_I \mu_N \mathbf{I} \cdot \mathbf{B}_{\text{ext}} .$$

Explain the physical origin of each of these terms, and discuss what is meant by a weak and strong external magnetic field in the context of hyperfine structure. [6]

Show that, in the weak field limit, a hyperfine level denoted by the quantum numbers J , I and F is split into states shifted in energy by

$$\Delta E_F = g_F \mu_B B_{\text{ext}} M_F ,$$

and obtain an expression for g_F in terms of g_J . [8]

Determine the value of g_J for the alkali atom ^{39}K in its ground state configuration $4s$. This atom has nuclear spin $I = 3/2$. Calculate the possible values of F , and the values of g_F associated with them. [4]

Atoms of ^{39}K are placed in a weak magnetic field. It is found that radio-frequency transitions with seven different frequencies can be induced between the hyperfine states of different values of F of the $4s$ configuration. By use of a diagram, sketch the energy levels and the values of F and M_F of the upper and lower states involved in the transitions associated with these seven frequencies. Mark on your sketch which transitions would be seen if observing along the direction of the magnetic field. [7]

3. The energy levels of a diatomic molecule can be written in the form

$$E = E_e(R_0) + (n + 1/2) h\nu_v + B K(K + 1) .$$

Clearly identify the physical origin of the terms appearing in this equation. [6]

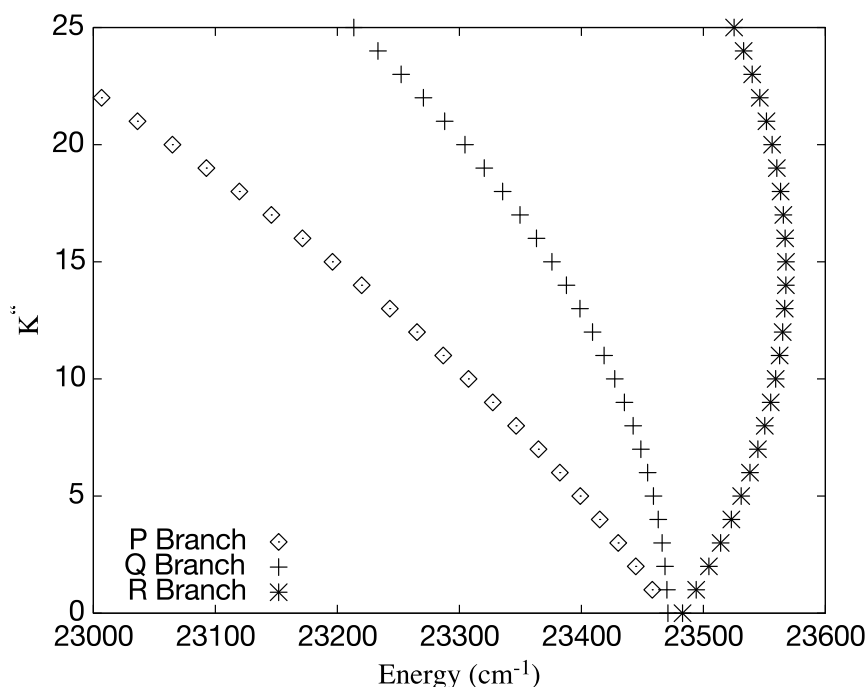
Molecules of ^{27}AlH are excited through an electrical discharge to levels of the $A^1\Pi$ electronic state by electrons colliding with molecules in the $X^1\Sigma^+$ ground electronic state. What is the total electron spin and electron orbital angular momentum along the internal axis associated with each of these states. [2]

The frequency of the transitions between the rotational levels of the lowest vibrational level of $A^1\Pi$ and the rotational levels of the lowest vibrational level of $X^1\Sigma^+$ are measured. Show that the frequencies of the P branch ($\Delta K = +1$ in emission) are given by

$$h\nu_P = h\nu_{00} - (B' + B'') K'' + (B' - B'') K''^2 ,$$

where B'' and K'' are associated with the lower electronic state, and B' is associated with the upper electronic state. Deduce the corresponding expressions for the Q and R branches. [6]

The figure below shows the measured energies (in wavenumbers) of the transitions for different values of K'' . In terms of B' and B'' , derive an expression for the value of K'' at which the band head forms. Separate measurements show that $B''/(hc) = 6.3907 \text{ cm}^{-1}$. Find ν_{00} and $B'/(hc)$ and calculate the equilibrium separation of the nuclei within the molecule for each of the two electronic levels. [11]



4. The energies of X-ray transitions in an atom of atomic number Z can be represented by the expression

$$E_{nm} = hcR_{\infty} \left[\frac{(Z - \sigma_n)^2}{n^2} - \frac{(Z - \sigma_m)^2}{m^2} \right] ,$$

where n and m are integers. Explain the form of this expression and the physical significance of the quantities σ_n and σ_m . [7]

Explain what transitions are involved in K_{α} radiation, and what is meant by the K -edge of an absorbing material. Show how their energies can be calculated from the above formula. Ca ($Z = 20$) emits K_{α} radiation at an energy of 3692 eV, and has a K absorption edge at 4039 eV. Calculate the values of σ for the two atomic levels associated with the K_{α} transition in Ca, and comment on their values. [6]

Show that K_{α} radiation from an element with atomic number Z will be efficiently absorbed by a filter made from the element with atomic number $Z - 1$ if

$$Z < \sigma_2 + 4 + \sqrt{(\sigma_2 + 4)^2 - (\sigma_2^2 + 8\sigma_1 + 4)} . \quad [6]$$

Assuming that the values of σ do not vary greatly as a function of Z , determine the element with the highest atomic number Z' whose K_{α} emission is efficiently absorbed by a filter made from element $Z' - 1$. For this situation calculate the difference between the K_{α} energy and the K -edge energy of the absorbing filter. [6]